

AMENDMENT NO. 1 JUNE 1994
TO
IS 1554 (Part 1) : 1988 SPECIFICATION FOR PVC
INSULATED (HEAVY DUTY) ELECTRIC CABLES
PART 1 FOR WORKING VOLTAGES UP TO AND
INCLUDING 1 100 V

(Third Revision)

(Page 1, clause 0.3) — Delete this clause and renumber the subsequent clauses.

(Page 1, clause 0.5) — Substitute the following for the existing clause:

‘0.4 A special category of cables with improved fire performance has been included in this standard. Classification of such cables is given in Appendix A.’

(Page 1, clause 1.1) — Insert the following new clause 1.2 after 1.1 and renumber the subsequent clauses:

‘1.2 This standard also covers cables with improved fire performance, categories C1 and C2, as given in Appendix A. For such cables additional requirements have been included wherever necessary in 7.1.3, 15.1.1, 15.2.1 and 17.2.2).

NOTE — Normal cables to this standard can be classified as meeting the requirements of category 01.’

[Page 2, clause 6.1(b)] — Substitute ‘formed wire’ for ‘strip’.

(Page 2, clause 6.2) — Replace first sentence by the following:

‘The galvanized steel wires/formed wires/tapes shall comply with the requirements of IS 3975 : 1988, except that the provisions of 3.3.1 and 3.3.2 of IS 3975 shall not apply.’

(Page 2, clause 7.1.2) — Insert the following new clause 7.1.3 after 7.1.2:

‘7.1.3 For cables with improved fire performance, the outer sheath shall, in addition, meet the requirement of tests applicable for the required category (see 15.1.1 and 15.2.1).

(Page 3, clause 8.1, Informal table, captions for col 1 and 2) — Substitute the following for the existing:

Nominal Cross Sectional Area	
Copper mm²	Aluminium mm²

(Page 4, clause 10.3.1, para 2) — Delete letter 'd' in line five; also delete the last sentence.

(Page 4, Fig. 1) — Delete.

(Page 6, clause 14.4.2) — Substitute the following for the existing:

'14.4.2 Armoured Cables — The thickness of outer sheath shall be not less than the minimum value specified in col 5 of Table 7.'

(Page 8, clause 15.1) — Insert the following new clause 15.1.1 after 15.1:

'15.1.1 The following shall constitute additional type tests for cables with improved fire performance as per the categories given in Appendix A:

Category	Tests	For require- ment, Ref	For Test Methods, Ref. Part No. of IS 10810
(1)	(2)	(3)	(4)
01	No Additional tests	—	—
C1	a) Oxygen index test	16.5	58 (Under preparation)
	b) Flame retardance test on single cable	16.6	61
	c) Flame retardance test on bunched cables	16.7	62
	d) Temperature index	16.10	Under consideration
C2	a) Oxygen index test	16.5	58 (Under preparation)
	b) Flame retardance test on single cable	16.6	61
	c) Flame retardance test on bunched cables	16.7	62

d) Test for specific optical density of smoke	16.8	Under Consideration
e) Smoke density	16.11	63
f) Test for halogen acid gas evolution	16.9	59
g) Temperature index	16.10	Under consideration

NOTES :

1. For category C1, tests (a) and (d) are to be performed on samples taken from outer sheath, as applicable, and prepared in the manner given in the relevant test method.
2. For category C2, tests (a), (e), (f) and (g) are to be performed on samples taken from outer sheath, as applicable, and prepared in the manner given in the relevant test method.

(Page 9, clause 15.2) — Add clause 15.2.1: and renumber the existing clause as 15.2.2.

‘15.2.1 The following shall constitute additional acceptance tests for cables with improved fire performance as per the categories given in Appendix A:

Category	Test
01	No additional Test
C1	a) Oxygen index test b) Flame retardance test on single cable
C2	a) Oxygen index test b) Flame retardance test on single cable c) Test for specific optical density of smoke d) Test for halogen acid gas evolution e) Test for smoke density

(Page 10, clause 16.4) — Add the following clauses:

‘16.5 Oxygen Index Test

The test on samples of inner/outer sheath (as applicable) shall be done at $27 \pm 2^{\circ}\text{C}$. The oxygen index shall not be less than 29.

16.6 Flame Retardance Test on Single Cables

After the test, there should be no visible damages on the test specimen within 300 mm from its upper end. Marks from fixing devices, soot or changing of the colour are not considered damages.

16.7 Flame Retardance Test on Bunched Cables

After burning has ceased, the cables should be wiped clean and the charred or affected portion should not have reached a height exceeding 2.5 m above the bottom edge of the burner, measured at the front and rear of the cable assembly.

NOTE — Requirements for this test are split in 3 categories i.e. A, B and C as described in IS 10810 (Part 62) : 1993. For the purpose of this standard, category B and C test methods shall be used. In the absence of any special requirements for method B, method C shall be used for both the categories C1 and C2.

16.8 Test for Specific Optical Density of Smoke

Under Consideration.

16.9 Test for Halogen Acid Gas Evolution

The level of HCl evolved shall not exceed 20 percent by weight.

16.10 Test for Temperature Index

The measured value of Temperature Index shall be 21 at a temperature of 250°C .

16.11 Smoke Density

Under consideration.’

(Page 10, clause 17.2.1) — Add new clause:

‘17.2.2 The following special cables shall be identified by indenting, embossing or printing the appropriate legend on the outer sheath throughout the cable length, in addition to the existing marking requirements.’

Type of Cable	Legend
i) Improved fire performance or Category C1	FR
ii) Improved fire performance for Category C2	FR—LSH

(Page 11, Appendix A) — Insert the following new 'Appendix A' and renumber the existing 'Appendix A' as 'Appendix B':

APPENDIX A

(*Clauses 0.4, 1.2 and 15.1.1*)

CLASSIFICATION OF CABLES FOR IMPROVED FIRE PERFORMANCE

Category	Environment Description	Type	Cable Definition
01	Cables in open areas	—	Flame retardent, Single cable self-extinguishing, does not propagate flame
C1	Cables in constrained areas	FR	Flame retardent, Does not propagate fire even when installed in groups in vertical ducts
C2	Cables in constrained areas with limited human activity and/or presence of sophisticated systems	FR—LSH	Flame retardent cables with reduced halogen evolution and smoke

(ETD 09)